Program 9: Creating MavenAnsibleWebApp

Step 1: This below Maven command creates a new ansible web application project named MavenAnsibleWebApp using the maven-archetype-webapp template. It sets the group ID to com.example and runs in non-interactive mode, generating all project files automatically.

mvn archetype:generate -DgroupId=com.example -DartifactId=MavenAnsibleWebApp DarchetypeArtifactId=maven-archetype-webapp -DinterativeMode=false

```
brijesh@Ubuntu:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyAnsibleWebApp1 -DarchetypeArtifactI
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/brijesh/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyAnsibleWebApp1
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/brijesh/devops/MyAnsibleWebApp1
[INFO] ----------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ----------------------------------------------------------------------------
[INFO] Total time:  2.814 s
[INFO] Finished at: 2026-05-10T17:13:47+05:30
[INFO] ----------------------------------------------------------------------------
brijesh@Ubuntu:~/devops$
```

Step 2: Configure Maven Ansible Web App Project (pom.xml) In this step, we move to the project directory using cd MavenAnsibleWebApp and open the Maven configuration file using gedit pom.xml. Then, we write the required project details, dependencies, WAR packaging settings, and build configuration for creating the web application.

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example</groupId>
<artifactId>MyAnsibleWebApp</artifactId>
<packaging>war</packaging>
<version>1.0-SNAPSHOT</version>
<name>MyAnsibleWebApp</name>
<url>http://maven.apache.org</url>
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>3.8.1</version>
<scope>test</scope>
</dependency>
<!--  Servlet API for Java EE  -->
<dependency>
<groupId>javax.servlet</groupId>
<artifactId>javax.servlet-api</artifactId>
<version>4.0.1</version>
<scope>provided</scope>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-war-plugin</artifactId>
<version>3.4.0</version>
<!--  Update to latest stable version  -->
```

```
</plugin>
</plugins>
<finalName>MyAnsibleWebApp</finalName>
<!--  create a web page with the name MavenAnsibleWebApp, if we want
some other name we can chage it  -->
</build>
</project>
```

Step 3: Create Ansible Configuration Files In this step, we create a new folder named ansible inside the project directory using mkdir ansible. Then, we move into the folder using cd ansible and create the required Ansible files using gedit hosts.ini and gedit playbook.yml. These files are used to define target hosts and deployment tasks for automation.

```
brijesh@Ubuntu:~/devops$ ls
MyAnsibleWebApp  MyAnsibleWebApp1  token
brijesh@Ubuntu:~/devops$ cd MyAnsibleWebApp
brijesh@Ubuntu:~/devops/MyAnsibleWebApp$ ls
ansible  Jenkinsfile  pom.xml  src  target
brijesh@Ubuntu:~/devops/MyAnsibleWebApp$ cd ansiblke
bash: cd: ansiblke: No such file or directory
brijesh@Ubuntu:~/devops/MyAnsibleWebApp$ cd ansible
brijesh@Ubuntu:~/devops/MyAnsibleWebApp/ansible$ ls
hosts.ini  playbook.yml
brijesh@Ubuntu:~/devops/MyAnsibleWebApp/ansible$
```

gedit host.ini

This command is written inside hosts.ini file to define the Ansible inventory. It creates a group named web and adds localhost as the target system, using local connection and Python 3 interpreter for executing Ansible tasks.

```
[web] localhost ansible_connection=local
ansible_python_interpreter=/usr/bin/python3
```

gedit playbook.yml

This code is written inside playbook.yml file to automate deployment using Ansible. It targets the web host group, copies the generated WAR file from the Jenkins workspace to the Tomcat webapps folder, and then

restarts the Tomcat service so the updated application runs successfully. - name:
Deploy to Tomcat hosts: web become: yes tasks:
- name: Copy WAR file to Tomcat copy:
src: "/var/snap/jenkins/5013/workspace/MavenAnsibleWebApp1 CICD/target/MavenAnsibleWebApp1.war"
dest: "/opt/tomcat/webapps/MavenAnsibleWebApp1.war"
remote_src: no - name: Restart Tomcat
systemd:
name: tomcat
state: restarted

Step 4: Build the Maven Ansible Web Application In this step, we run the command mvn clean install to clean previous files and build the project. Maven compiles the application, packages it as a .war file, and stores it in the target folder. After successful completion, the build shows BUILD SUCCESS and installs the project files into the local Maven repository.

```
brijesh@Ubuntu:~/devops/MyAnsibleWebApp$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:MyAnsibleWebApp >--------------------
[INFO] Building MyAnsibleWebApp 1.0-SNAPSHOT
[INFO] --------------------------------[ war ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyAnsibleWebApp ---
[INFO] Deleting /home/brijesh/devops/MyAnsibleWebApp/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyAnsibleWebApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MyAnsibleWebApp ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyAnsibleWebApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/brijesh/devops/MyAnsibleWebApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ MyAnsibleWebApp ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ MyAnsibleWebApp ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (default-war) @ MyAnsibleWebApp ---
[INFO] Packaging webapp
[INFO] Assembling webapp [MyAnsibleWebApp] in [/home/brijesh/devops/MyAnsibleWebApp/target/MyAnsibleWebApp]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/brijesh/devops/MyAnsibleWebApp/src/main/webapp]
[INFO] Building war: /home/brijesh/devops/MyAnsibleWebApp/target/MyAnsibleWebApp.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyAnsibleWebApp ---
[INFO] Installing /home/brijesh/devops/MyAnsibleWebApp/target/MyAnsibleWebApp.war to /home/brijesh/.m2/repository/com/example/MyAnsibleWebApp/1.0-SNAPSHOT/MyAnsibleWebApp-1.0-SNAPSHOT.war
[INFO] Installing /home/brijesh/devops/MyAnsibleWebApp/pom.xml to /home/brijesh/.m2/repository/com/example/MyAnsibleWebApp/1.0-SNAPSHOT/MyAnsibleWebApp-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.413 s
[INFO] Finished at: 2026-05-10T17:48:26+05:30
[INFO] ------------------------------------------------------------------------
brijesh@Ubuntu:~/devops/MyAnsibleWebApp$
```

Step 5: Verify Project Structure In this step, we run the tree command to display the complete project structure. It shows important files and folders such as pom.xml, Jenkinsfile, ansible configuration files, source code folders, and the target directory containing the generated MavenAnsibleWebApp1.war file ready for deployment.

```
brijesh@Ubuntu:~/devops/MyAnsibleWebApp$ tree
.
├── ansible
│   ├── hosts.ini
│   └── playbook.yml
├── Jenkinsfile
├── pom.xml
├── src
│   └── main
│       ├── resources
│       └── webapp
│           ├── index.jsp
│           └── WEB-INF
│               └── web.xml
└── target
    ├── classes
    ├── maven-archiver
    │   └── pom.properties
    ├── MyAnsibleWebApp
    │   ├── index.jsp
    │   ├── META-INF
    │   └── WEB-INF
    │       ├── classes
    │       └── web.xml
    └── MyAnsibleWebApp.war

13 directories, 10 files
brijesh@Ubuntu:~/devops/MyAnsibleWebApp$
```

Setting Up Jenkins Pipeline for MavenAnsibleWebApp

Step 1: To deploy MavenAnsibleWebApp, open the terminal, cd into the MavenAnsibleWebApp folder, create a Jenkinsfile containing the pipeline code, and then commit and push it to GitHub.

```
pipeline {
    agent any  // Use any available agent

    environment {
        LANG = 'en_US.UTF-8'
        LC_ALL = 'en_US.UTF-8'
    }   // this has to be added only if you get an error saying UTF required is
8 but showing in ISO00009

    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url:
'https://github.com/Brijesh-ue/MyAnsbileWebApp.git'
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean package'  // Run Maven build
            }
        }

     stage('Archive') {
            steps {
                archiveArtifacts artifacts: 'target/*.war', fingerprint:true
            }
        }
        stage('Deploy') {
            steps {
```

```
            sh 'mvn clean package'
            sh 'ansible-playbook ansible/playbook.yml -i ansible/hosts.ini'
        }
    }



  }

 }
```

Step 2: To deploy MavenAnsibleWebApp, you first need to create a Jenkins pipeline job that will use the Jenkinsfile from your project repository.

Then, go to the Jenkins home page and click on “New Item” to create a job, enter the item name (e.g., MavenAnsibleWebApp), and select Pipeline as the project type.

This option allows you to build, test, and deploy applications using a Jenkinsfile. After selecting the pipeline type, click “OK” to proceed to the job configuration page.

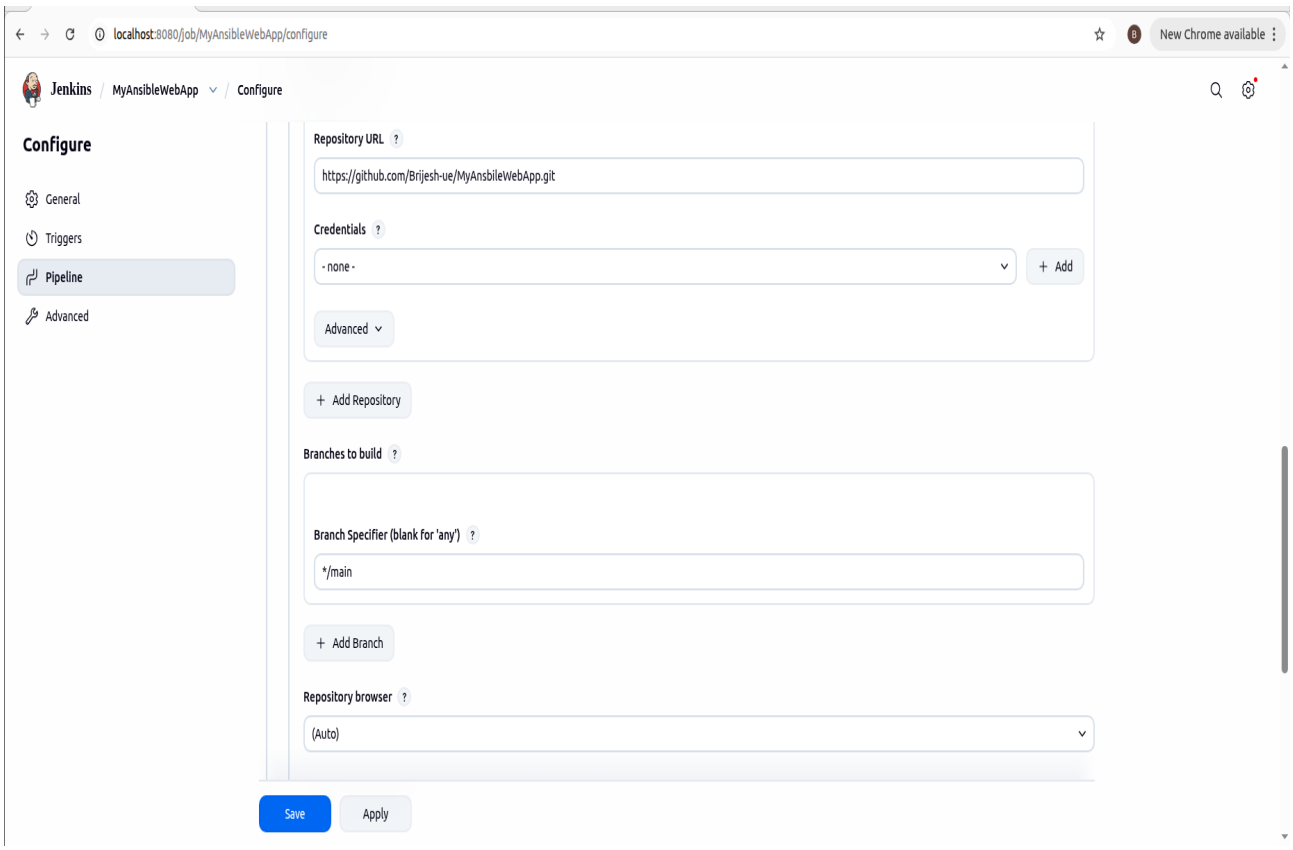
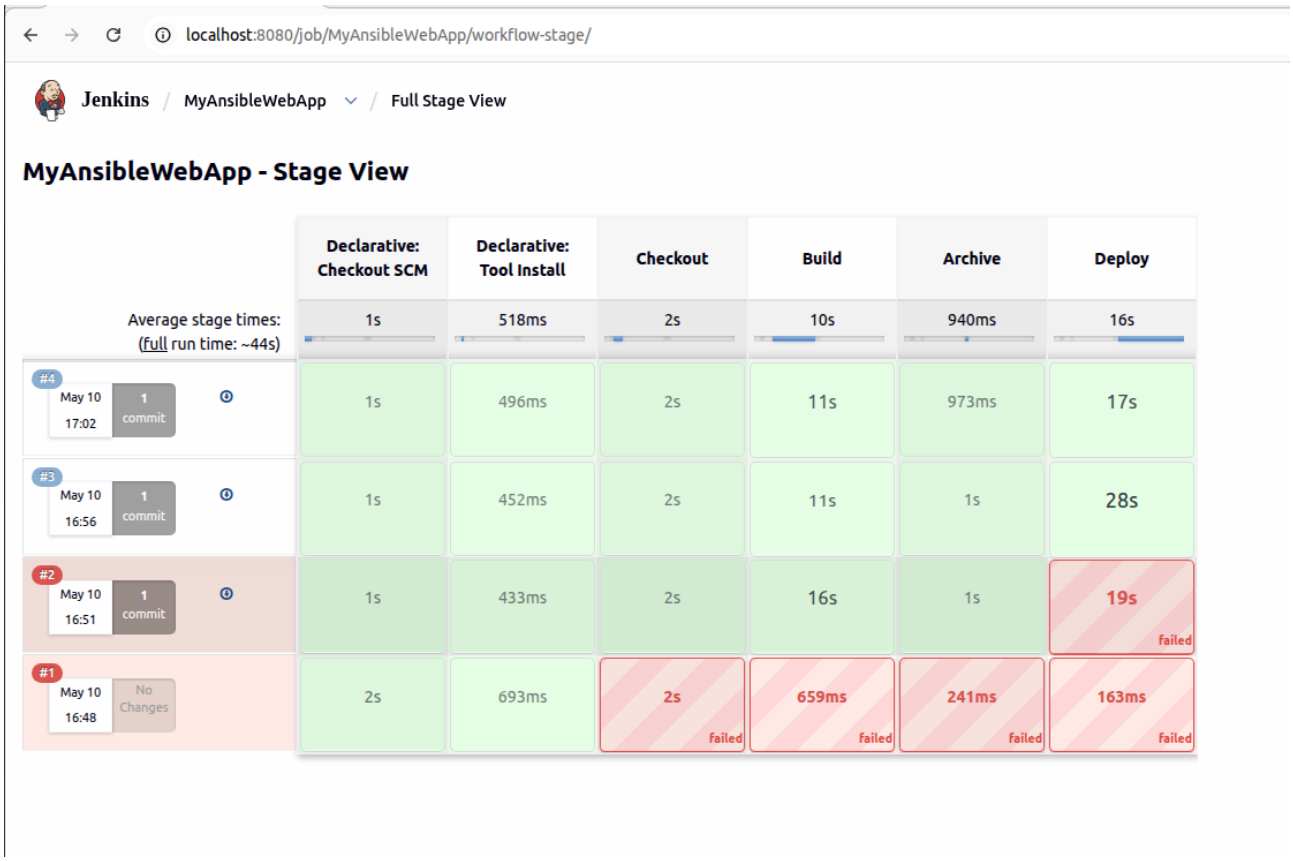
Step 3: In MavenAnsibleWebApp, go to the Pipeline section and under Definition, select Pipeline script from SCM.

Choose Git as the SCM, and under Repository URL, paste the GitHub URL of MavenAnsibleWebApp.

In the Credentials field, select the previously created credential (e.g., GitHub Credential).

Finally, in Script Path, enter the name of the Jenkinsfile you created in the MavenAnsibleWebApp folder and pushed to GitHub.

Then click Apply and Save to finalize the configuration.



Step 4: Click on Build Now and then Full Stage View to view the pipeline output